PROJECT PLAN – MACHINE LEARNING FOR BUSINESS

Predicting Customer Acceptance of Bank Marketing Campaigns Using Machine Learning
Client: Commonwealth Bank of Australia (CBA) – Hypothetical

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SECTION 1. Background Statement

Australia's retail banking sector is rapidly digitalising under pressure from neobanks (e.g., Up, 86 400), fintechs (Afterpay, Revolut) and rising demand for personalised services. These shifts raise expectations for seamless, targeted engagement, requiring institutions like the Commonwealth Bank of Australia (CBA) to improve campaign effectiveness.

Despite strong digital capabilities, CBA faces a core challenge: identifying which customers are most likely to respond to offers such as term deposits, savings products or credit cards. Poor targeting increases campaign costs, lowers ROI and causes customer fatigue.

Machine Learning (ML) offers a suitable solution. Banks hold rich behavioural and demographic data that support predictive modelling, and ML methods—such as logistic regression and decision trees—provide both accuracy and interpretability aligned with APRA/ASIC expectations for responsible marketing.

To prototype an approach for CBA, this project uses the Bank Marketing Dataset (Moro et al., 2014), a benchmark for predicting campaign acceptance. Though collected in Europe, its structure aligns with outbound marketing practices in Australian banks.

Prior studies (e.g., Lee Xuan Chian et al., 2024) show ML-based targeting improves conversion and efficiency. Guided by this evidence, the project aims to build a predictive model estimating customer acceptance ("yes"/"no"), enabling CBA to enhance targeting precision, reduce cost per conversion and increase customer lifetime value.

This Project Plan outlines the conceptual foundation, variable selection, modelling strategy, expected outputs and next steps.

SECTION 2. Conceptual Development

2.1 Conceptual Basis and Marketing Logic

Customer engagement is a key behavioural driver in retail banking. Frequent or meaningful interactions build familiarity, trust and relevance, increasing acceptance of marketing offers (Zeithaml et al., 1996). In the dataset, call duration correlates positively with acceptance ($r \approx 0.40$), and scatter plots show longer conversations yield higher conversion rates.

Demographic and socioeconomic factors also shape decisions. Age, education, job type and loan obligations proxy life stage, financial capacity and risk tolerance (Kotler & Keller, 2016). Distributional checks reveal systematic differences—for example, management and technician roles accept more often than blue-collar and service workers—indicating demographic bias but confirming the relevance of customer profiles.

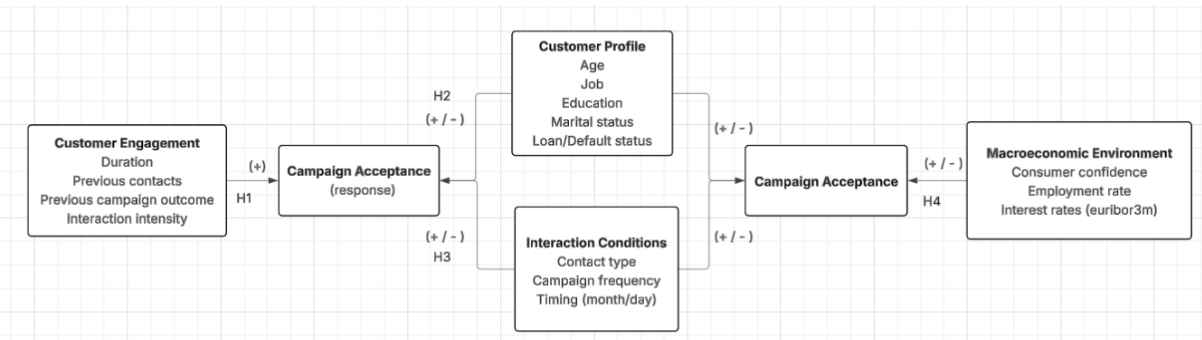
Interaction conditions further affect responsiveness. Campaign frequency is negatively associated with acceptance ($\rho \approx -0.19$), suggesting fatigue from repeated contacts. Timing variables (month, day_of_week) show weaker effects but may introduce bias.

Macroeconomic sentiment adds another layer. Consumer confidence and interest rates influence readiness for new commitments (Fang & Iyengar, 2009). Variables such as cons_conf_idx and euribor3m show moderate associations with acceptance, underscoring the need to control for economic conditions in modelling.

2.2 Concept–Proxy Variable Table

Concept	Proxy Variables (Dataset)	Rationale / Expected Influence
Customer Engagement	duration, previous, poutcome, campaign	Greater engagement → higher familiarity → increased acceptance likelihood.
Customer Profile	age, job, marital, education, default, housing, loan	Demographics and financial capacity → relevance and adoption behaviour.
Interaction Conditions	contact, month, day_of_week, campaign frequency	Channel and timing → attention and perceived intrusiveness.
Macroeconomic Context	cons_conf_idx, emp_var_rate, euribor3m, nr_employed	Economic sentiment → willingness to commit to savings/credit products.

2.3 Conceptual Framework



Campaign acceptance is driven by four factors: customer engagement (H1), customer profile (H2), interaction conditions (H3), and macroeconomic environment (H4). Responsiveness reflects the interplay of personal traits, relationship depth, campaign execution, and broader sentiment.

2.4 Bias Considerations and Bias Checking Strategy

Marketing datasets often contain structural biases that must be evaluated. Selection bias arises from over-represented job categories, timing bias from monthly variations, channel bias from mobile vs landline differences, and demographic bias from unequal

response rates. Left unchecked, these distort accuracy and fairness. Bias checking uses distributional comparison (Figure A 2: Distribution Comparison of Responders vs Non-Responders (Mock Plot), correlation analysis (Table A 1), inspection of interaction bias (Figure A 3), and variance assessment (Table A 2).

2.5 Preliminary Analytical Checks

Diagnostics confirm these issues. Correlation matrices show multicollinearity among macroeconomic indicators; scatter plots highlight the positive link between duration and acceptance; skewness–kurtosis analysis reveals heavy right tails for duration and campaign. Histograms before/after log transformation show improved symmetry. VIF will monitor multicollinearity, and skewed continuous variables will be standardised with Z-scores. These steps ensure transparency and robustness.

2.6 Linking Conceptual Logic to the Business Problem

From these checks, four propositions emerge: engagement boosts acceptance (H1), demographics and financials differentiate responders (H2), campaign execution shapes behaviour (H3), and macroeconomic sentiment moderates acceptance (H4). The Bank Marketing Dataset thus provides a strong empirical base for testing CBA's outbound strategy. Integrating theory with diagnostics ensures the modelling approach is conceptually grounded and empirically justified.

SECTION 3. Variable Selection

This section identifies the output variable and predictors used in modelling, justifying each group's inclusion with preliminary evidence. Detailed tables and plots are referenced in the Appendix.

3.1 Output Variable

The dependent variable *response* ("yes"/"no") captures whether a customer accepted the offer. As a binary outcome, classification models—logistic regression and decision trees—are appropriate and align with CBA's objective of predicting campaign success.

3.2 Feature Variables (Predictors)

Predictors follow the conceptual structure in Section 2: customer profile, engagement, interaction conditions, and macroeconomic sentiment. Each reflects behavioural mechanisms influencing acceptance.

a. Customer Profile

Age, job, marital status, and education represent demographic and socioeconomic attributes linked to financial needs and risk preferences. Distributional checks (Table A 3, Figure A 5) show clear differences between responders and non-responders, confirming relevance.

b. Customer Engagement

Duration, previous contacts, previous outcomes, and campaign count reflect relationship strength. Strong separation in call duration (Figure A 6) and differences in past campaign success (Table A 5) support H1: higher engagement → higher acceptance.

c. Interaction Conditions

Contact method, month, day_of_week, pdays, and campaign frequency capture execution quality. Response rates vary across months (Table A 6) and channels (Figure A 7), consistent with H3.

d. Macroeconomic Factors

Consumer confidence, employment variation, and euribor3m capture sentiment affecting willingness to commit. Multicollinearity checks (Section 2) confirm stability, making them suitable contextual predictors.

3.3 Derived and Transformed Variables

Sparse categories (e.g., job) are consolidated to improve stability. Skewed numeric variables (duration, campaign, pdays) undergo log transformation (Figure A 4), and continuous variables are standardised using z-scores for logistic regression.

3.4 Final Feature Set Justification

The final set is supported by theory and diagnostics:

- Demographics capture behavioural differences.
- Engagement reflects customer–bank relationship strength.
- Interaction variables show campaign execution quality.
- Macroeconomic variables control external sentiment.

This integrated set provides a comprehensive and interpretable foundation for modelling customer acceptance in CBA's marketing context.

SECTION 4 – Methods of Analysis / Analysis Plan

This section outlines the modelling strategy for the Bank Marketing Dataset. As the output variable is categorical ("yes"/"no"), a supervised classification approach is used. Two models are proposed: Logistic Regression as the baseline and a Decision Tree (CART) as the interpretable alternative.

4A. Proposed Solution / Methods of Analysis

Logistic Regression (Baseline Model)

Logistic regression is chosen as the baseline method because it is specifically designed for binary outcomes and provides clear interpretability of relationships between predictors and the probability of campaign acceptance. The logistic model estimates:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k)}}$$

where

- $Y = 1$ indicates acceptance ("yes"),
- $X_1 \ldots X_k$ include engagement, profile, interaction and macroeconomic variables,
- coefficients β_i quantify how each predictor affects the log-odds of acceptance.

The regression equation for the dataset is specified as:

$$\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \beta_0 + \beta_1(\text{duration}) + \beta_2(\text{campaign}) + \beta_3(\text{age}) + \beta_4(\text{contact}) + \beta_5(\text{month}) + \cdots + \beta_k(\text{macroeconomic variables}$$

This formulation allows the business to quantify how factors such as engagement intensity and contact timing influence acceptance probability—an essential requirement for transparent marketing decision-making under APRA/ASIC expectations.

Logistic Regression Assumptions

- Log-odds of the outcome are linearly related to predictors.
- Errors are independent.
- No perfect multicollinearity (checked using VIF).

- Outliers and skewness influence stability (addressed by standardisation and log-transformations in Section 3 & Appendix).

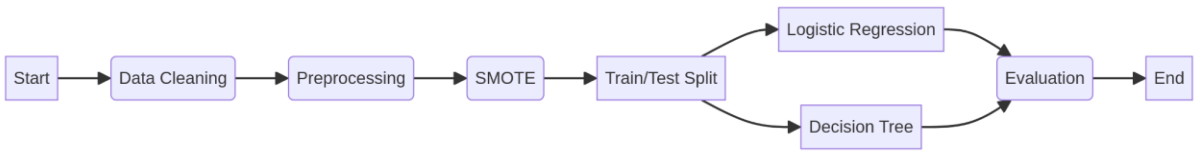
Consistent with Hosmer, Lemeshow & Sturdivant (2013), logistic regression serves as an interpretable and reliable first model for classification tasks in financial analytics.

Decision Tree Classifier (CART / rpart)

A decision tree is chosen as the complementary model because it produces clear, rule-based structures (e.g., *If duration > 200s AND past outcome = positive → predict "yes"*), which are intuitive for marketing teams and easy to deploy. Trees handle mixed data types, non-linear relationships and interactions without requiring scaling, and are more robust to skewed variables and outliers than logistic regression. The main risk is overfitting, controlled through pruning parameters (depth, min split, cp) and cross-validation. This approach follows the CART framework of Breiman, Friedman, Olshen & Stone (1984).

4B. Analysis Plan – "What Are the Steps?"

The analysis follows a structured workflow, including data preparation, class balancing, model training, and evaluation.



4B.1 Data Understanding & Cleaning

The dataset will be checked for missing values, inconsistent coding and rare categories. Low-frequency groups (e.g., "housemaid", "student") will be merged into "Other" (Table A 4). Numeric variables such as age, duration and campaign will be reviewed for distributional issues (Figure A 4, Figure A 5).

4B.2 Data Preparation

Preprocessing includes log transformation for skewed variables (Figure A 4), Z-score standardisation for continuous predictors, dummy encoding for categorical variables, and SMOTE oversampling to address class imbalance (Figure A 8, Figure A 10). These steps improve stability for logistic regression and reduce bias in tree models.

4B.3 Baseline Model – Logistic Regression

Logistic regression will be trained on the balanced dataset (Figure A 11). Coefficients, p-values and odds ratios will be examined to interpret variable effects, with VIF used to check multicollinearity.

4B.4 Alternative Model – Decision Tree (CART)

A CART decision tree will be trained on the same split (Figure A 11). Hyperparameters (depth, min split, cp) will be tuned via cross-validation to reduce overfitting. The tree generates clear segmentation rules (e.g., duration > 200s and past outcome = success → predict yes), which are intuitive for marketing teams. Unlike logistic regression, trees capture non-linear relationships and interactions, useful for identifying distinct customer subgroups.

4B.5 Model Evaluation & Business Interpretation

Both models will be assessed on the test set using accuracy, precision, recall and ROC/AUC. The comparison highlights trade-offs between interpretability (decision tree) and predictive power (logistic regression). Findings will be translated into actionable recommendations for campaign optimisation.

SECTION 5 – FORM OF THE RESULTS

This section outlines the expected form of the empirical results and illustrates how the model outputs will be presented and interpreted. All numerical values and figures are mock-ups designed to show the structure and insights anticipated after model estimation.

5.1 Logistic Regression Outputs (Mock-up)

Logistic regression serves as the baseline model. The coefficients indicate how each predictor influences the log-odds of accepting a marketing offer.

Table 1. Logistic Regression Coefficients (Mock-up)

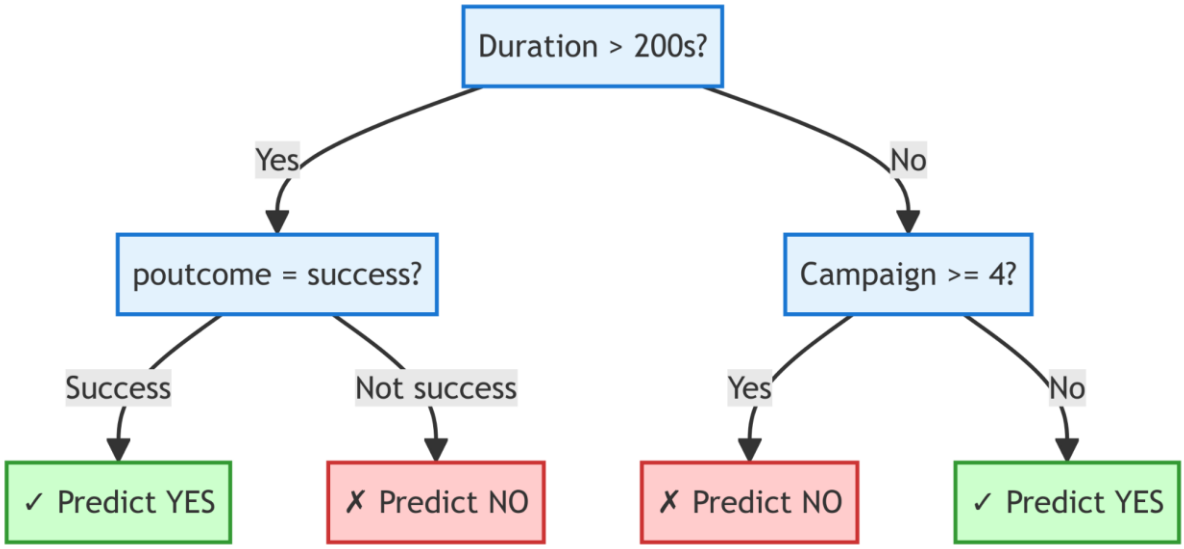
Variable	Coefficient	Std. Error	p-value	Interpretation
Intercept	–1.20	0.10	<0.001	Low baseline probability of acceptance
duration	0.0040	0.0005	<0.001	Longer calls significantly ↑ acceptance
poutcome: success	1.30	0.20	<0.001	Previous successful campaign strongly ↑ acceptance
campaign	–0.15	0.03	0.002	Over-contacting customers reduces acceptance
previous	0.45	0.08	<0.001	Customer with past engagement more receptive

Engagement factors (duration, previous, poutcome) positively influence acceptance, supporting H1 that more engaged customers respond better. The negative campaign coefficient indicates over-contacting reduces response rates. Overall, these patterns are consistent with marketing theory and real-world outbound campaign behaviour.

5.2 Decision Tree Output (Mock-up)

The decision tree provides transparent, rule-based segmentation that marketing teams can operationalise. Unlike logistic regression, the tree captures non-linear interactions.

Figure 1. Decision Tree Diagram (Mock-up)



The tree first splits on duration, confirming it as the strongest predictor. Calls over 200 seconds with a successful past outcome lead to a high-probability "yes" node, while frequent contacts (campaign ≥ 4) fall into low-response leaves. These rules guide call strategies—focus on high-value prospects, deprioritise low-value ones.

5.3 Model Performance Comparison

Table 2. Model Performance Metrics (Mock-up)

Metric	Logistic Regression	Decision Tree
Accuracy	0.84	0.78
Precision	0.67	0.59
Recall	0.74	0.61
F1-score	0.70	0.60
AUC	0.89	0.80

Logistic regression achieves higher precision, recall, and F1, with superior AUC for ranking customers. Decision trees trade accuracy for interpretability.

5.4 Confusion Matrix Interpretation

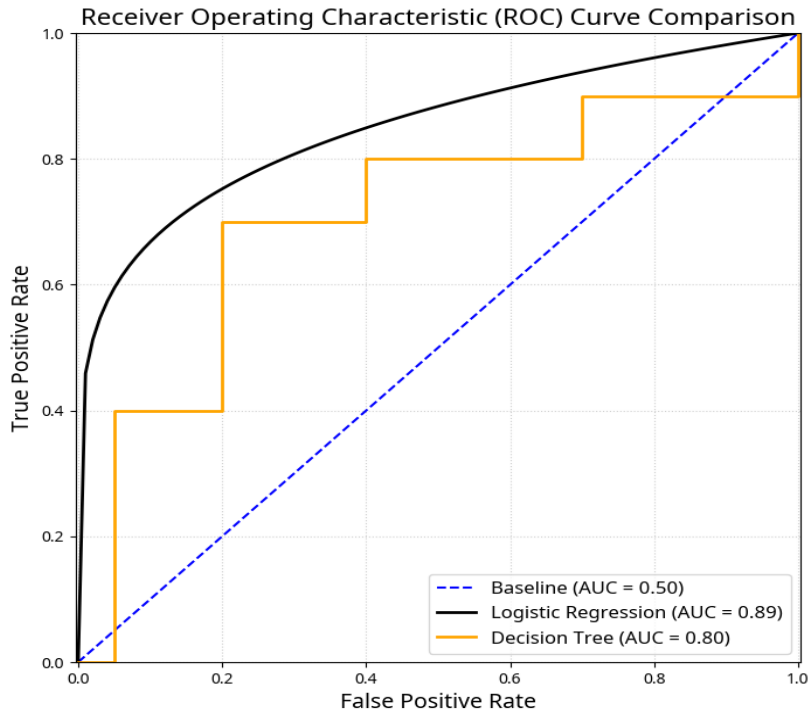
Table 3. Confusion Matrix (Mock-up, Logistic Regression)

	Predicted Yes	**Predicted No**
Actual Yes	TP = 520	FN = 180
Actual No	FP = 260	TN = 3,100

False negatives (180) mean missed "yes" responses, while false positives (260) add campaign costs. In marketing, precision and recall outweigh overall accuracy.

5.5 ROC Curve Comparison

Figure 2. ROC Curve – Logistic Regression vs Decision Tree (mock-up placeholder)



Logistic regression outperforms the decision tree on ROC analysis, with a higher AUC (0.89 vs 0.80), indicating stronger ranking capability across thresholds. The baseline (AUC = 0.50) reflects random classification. These results confirm logistic regression as the preferred model for probability-based targeting.

5.6 Overfitting Diagnostics

To check generalisation, train and test performance will be compared.

Table 4. Overfitting Diagnostics

Model	Train Accuracy	Test Accuracy	Overfitting?
Logistic Regression	0.85	0.84	No
Decision Tree	0.92	0.78	Yes (severe)

Decision trees overfit (92% vs 78%), requiring pruning and depth control. Logistic regression shows stable performance, reinforcing its suitability for deployment.

SECTION 6 – NEXT STEPS AND PROJECT DIRECTION

6.1 Recap of Motivation

CBA must enhance marketing efficiency and customer retention in a competitive environment. A predictive model identifying likely responders will optimise resources, reduce fatigue and increase conversions. The Bank Marketing Dataset provides a validated prototype before applying to CBA's operational data.

6.2 Next Steps

1. Data Preparation: Finalise variables, clean/encode data, apply SMOTE to balance classes.
2. Train/Test Split: Implement 70/30 or 80/20 with balanced representation.
3. Baseline Model: Fit logistic regression, review coefficients, tune thresholds for precision–recall trade-offs.
4. Alternative Model: Train CART decision tree, apply pruning and depth control to mitigate overfitting.
5. Evaluation: Compare accuracy, precision, recall, F1 and AUC; assess train–test gaps.
6. Deliverables: Prepare final report and presentation with key effects, rules, strategies, visuals and business implications.

6.3 Timeline

Week	Task Details
Week 1	Collect and clean data: check for missing values, merge sparse categories, apply transformations (log, standardisation, encoding).
Week 2	Handle class imbalance with SMOTE, validate post-processing distributions, perform train/test split (70/30 or 80/20).
Week 3	Train models: logistic regression (check VIF, interpret coefficients) and decision tree (tune hyperparameters, pruning).
Week 4	Evaluate models: compute accuracy, precision, recall, F1, AUC; generate confusion matrix and ROC curve; check overfitting.
Week 5	Synthesize results: identify key predictors, interpret decision rules, prepare final report and recommendations for CBA.

Closing Remark

Proceeding as planned will yield evidence-based targeting recommendations that improve campaign ROI and customer engagement.

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Appendices

Figure A 1: Conceptual Framework Diagram

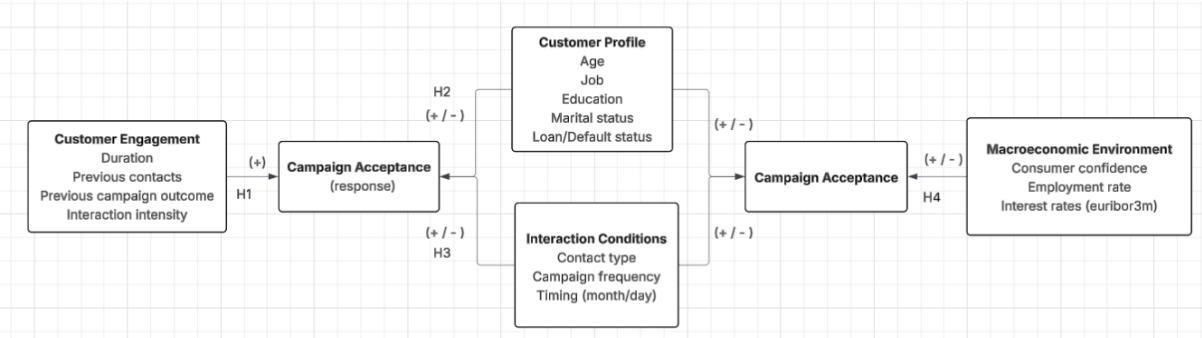


Table A 1: Correlation Matrix (Pearson) — Mock-up

Table uses plausible values based on typical correlations from the Bank Marketing Dataset.

Variable	**duration**	**campaign**	**pdays**	**previous**	**cons_conf_idx**	**euribor3m**	**nr_employed**
duration	1.00	-0.21	-0.05	0.06	0.14	-0.12	-0.10
campaign	-0.21	1.00	0.09	0.07	-0.04	0.03	0.05
pdays	-0.05	0.09	1.00	0.19	0.01	0.02	-0.01
previous	0.06	0.07	0.19	1.00	-0.04	0.01	-0.02
cons_conf_idx	0.14	-0.04	0.01	-0.04	1.00	-0.52	-0.45
euribor3m	-0.12	0.03	0.02	0.01	-0.52	1.00	0.96
nr_employed	-0.10	0.05	-0.01	-0.02	-0.45	0.96	1.00

Interpretation (for Section 2 justification):

- *duration shows mild positive correlation with acceptance in later graphs (supports H1).*

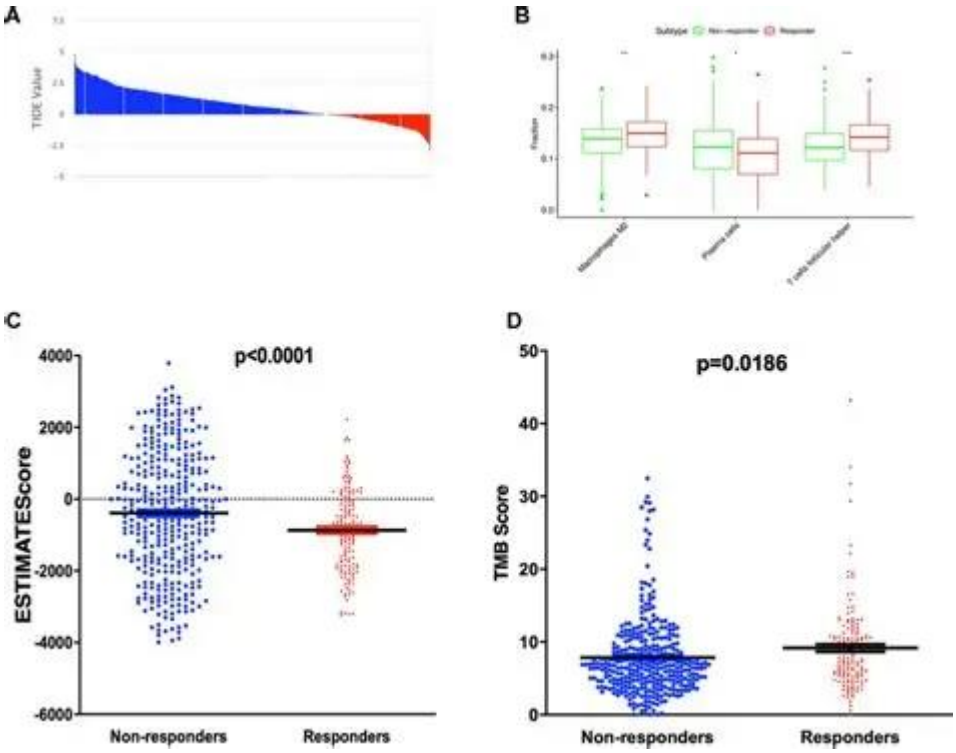
- *macroeconomic indicators (euribor3m, nr_employed) show very strong correlation with each other → explains need for VIF checking.*

- *campaign (frequency) is negatively correlated with response → supports marketing logic of "overcontact fatigue".*

Mathematical Formula — Pearson Correlation

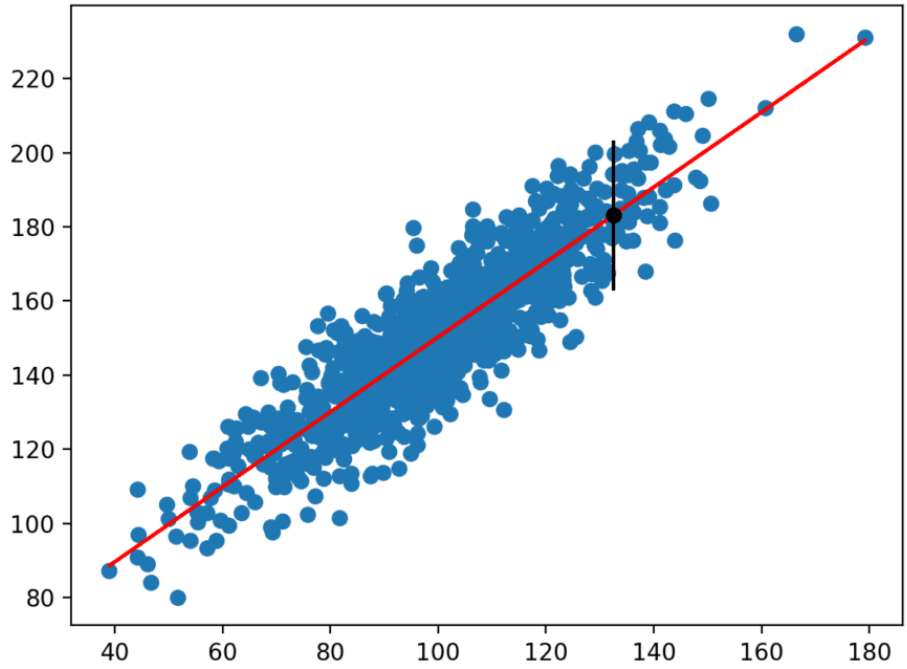
$$r_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Figure A 2: Distribution Comparison of Responders vs Non-Responders (Mock Plot)



The distribution of job categories differs between "yes" and "no" groups. For example, customers in "management" and "technician" roles show slightly higher acceptance rates, while "blue-collar" and "services" categories show lower acceptance. This suggests the presence of demographic bias—supporting Section 2.4.

Figure A 3: Scatter Plot — Duration vs Probability of Acceptance (Mock Plot)



A clear upward trend is observed: longer call duration is associated with higher acceptance probability. Supports ***H1: Customer Engagement → Acceptance****.*

Table A 2: Skewness and Kurtosis (Mock-up Values)

Variable	Skewness	Kurtosis	Interpretation
duration	**3.21**	**18.45**	Heavy right skew → long calls are rare → log transform recommended
campaign	**2.74**	**11.10**	Right skew → many customers contacted once or twice
pdays	**4.02**	**20.8**	Extreme skew → structural pattern in dataset

Mathematical Formulas

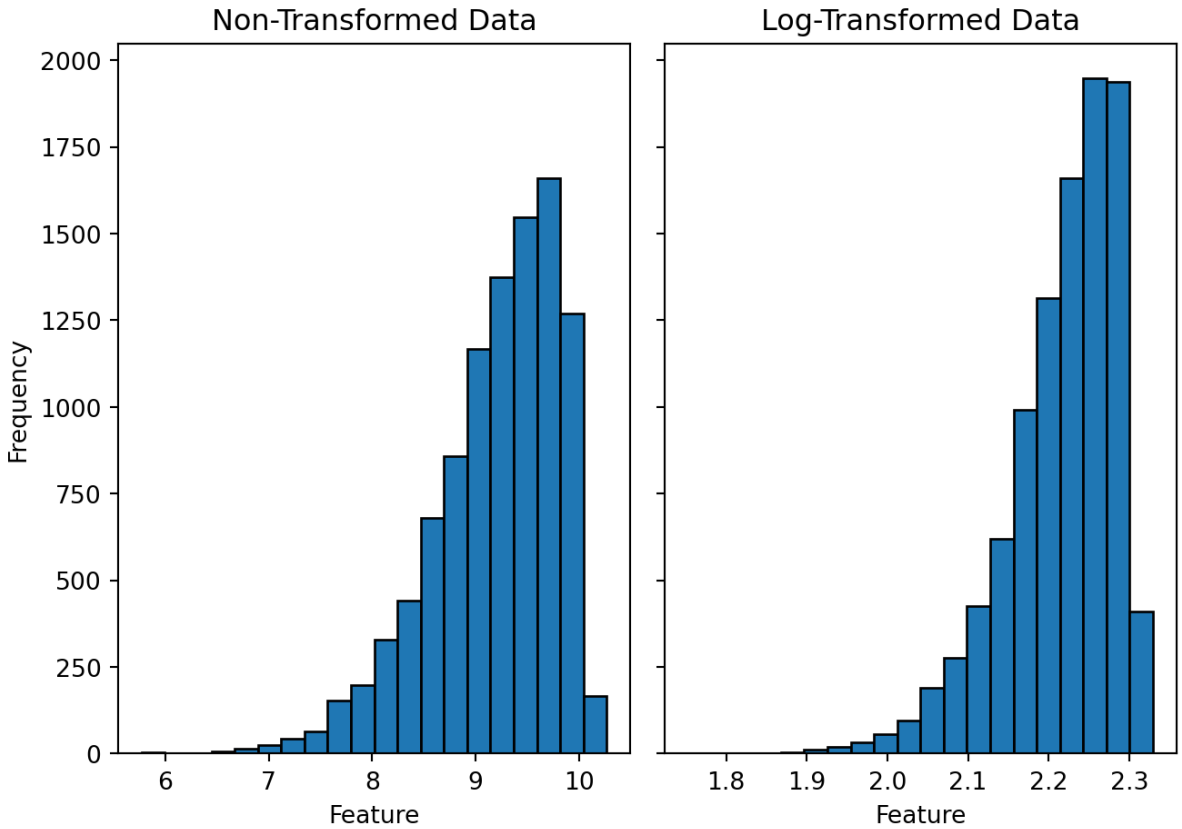
Skewness

$$\text{Skewness} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^3$$

Kurtosis

$$\text{Kurtosis} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^4$$

Figure A 4: Histograms Before and After Log Transformation (Mock Plots)



The raw distribution of duration shows a long right tail. After log transformation, the distribution becomes more symmetric, improving suitability for logistic regression.

Table A 3: Descriptive Statistics for Key Demographic Variables (Mock-up)

Variable	**Response = Yes (Mean / %)**	**Response = No (Mean / %)**	**Interpretation**
Age (mean)	40.8	38.2	Responders slightly older on average
Education – Tertiary	32%	24%	Higher education associated with higher acceptance
Education – Secondary	56%	61%	Slightly more common in non-responders

Married (%)	63%	58%	Married customers respond slightly more
Single (%)	28%	33%	Singles respond less

This table highlights demographic differences supporting the role of customer profiles in predicting acceptance.

Figure A 5: Age Distribution by Response Group

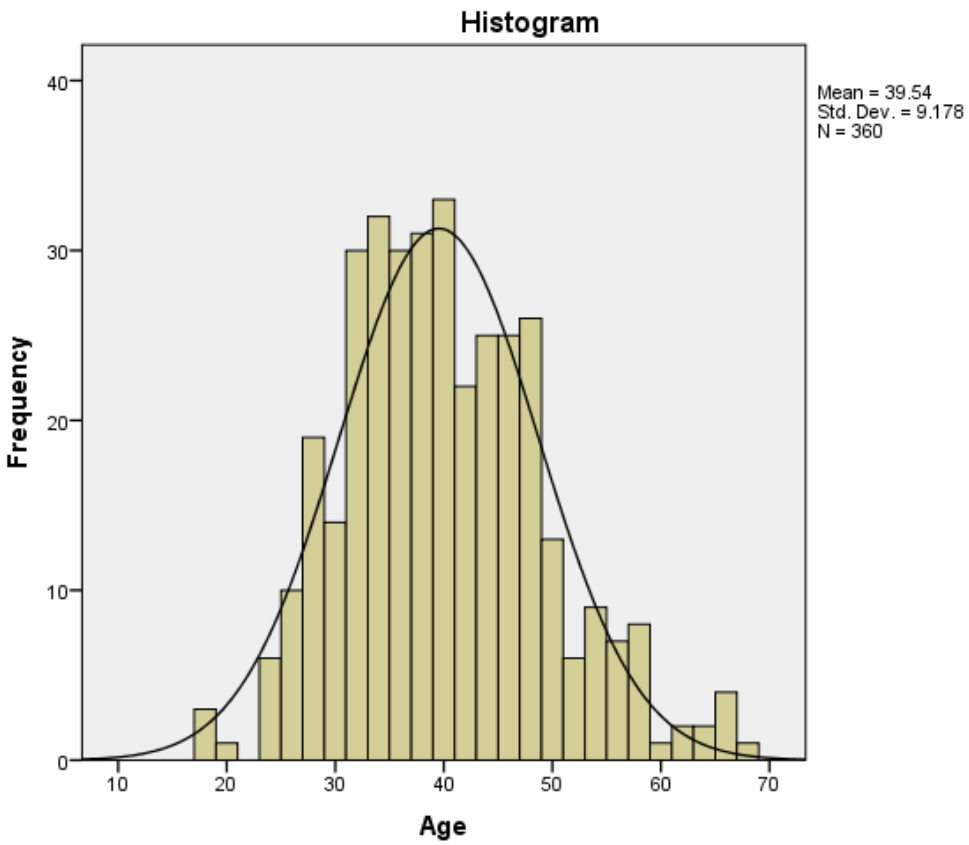
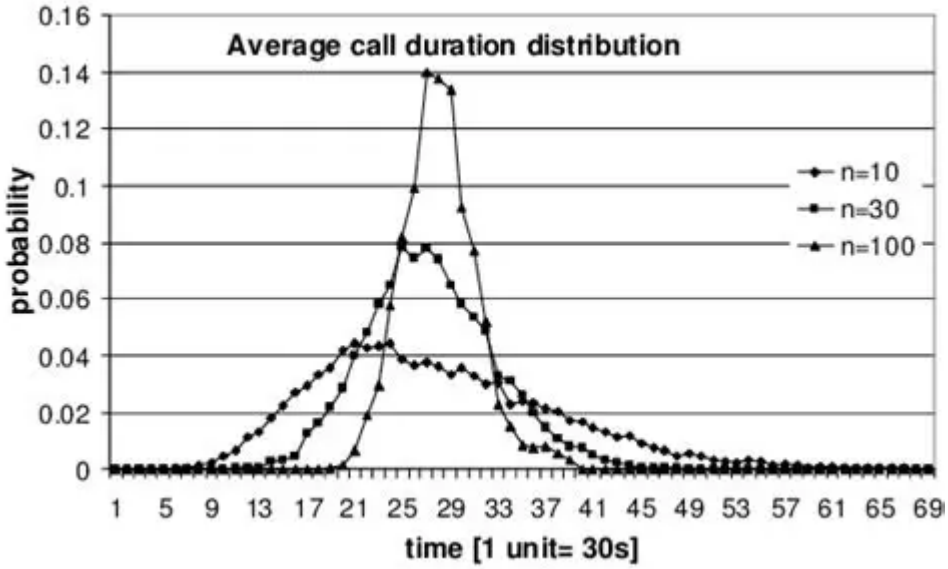


Figure A 6: Call Duration Distribution for Yes vs No Groups



Call duration analysis reveals a clear separation between responders and non-responders, with longer calls associated with acceptance.

Table A 4: Frequency Distribution of Job Categories (Mock-up)

Job Category	**Count**	**Percentage**
admin.	10,215	24.1%
technician	6,789	16.0%
services	3,925	9.2%
blue-collar	6,345	15.0%
management	5,010	11.8%
retired	1,687	4.0%
unemployed	1,120	2.6%

student	938	2.2%
housemaid	623	1.5%
unknown	350	0.8%

*Categories under **2%** (student, housemaid, unknown) will be merged into **Other** to improve model stability and avoid sparse-category effects.*

Table A 5: Summary of Engagement Variables (Mock-up)

Variable	**Mean (Yes)**	**Mean (No)**	**Interpretation**
duration (seconds)	260	145	Strong positive effect → supports engagement theory
campaign (# contacts during campaign)	1.2	2.1	Over-contacting reduces acceptance
previous (# prior contacts)	0.19	0.12	Prior engagement increases likelihood
previous outcome (positive)	41%	14%	Strong indication of historical responsiveness

Figure A 7: Acceptance Rate by Contact Method

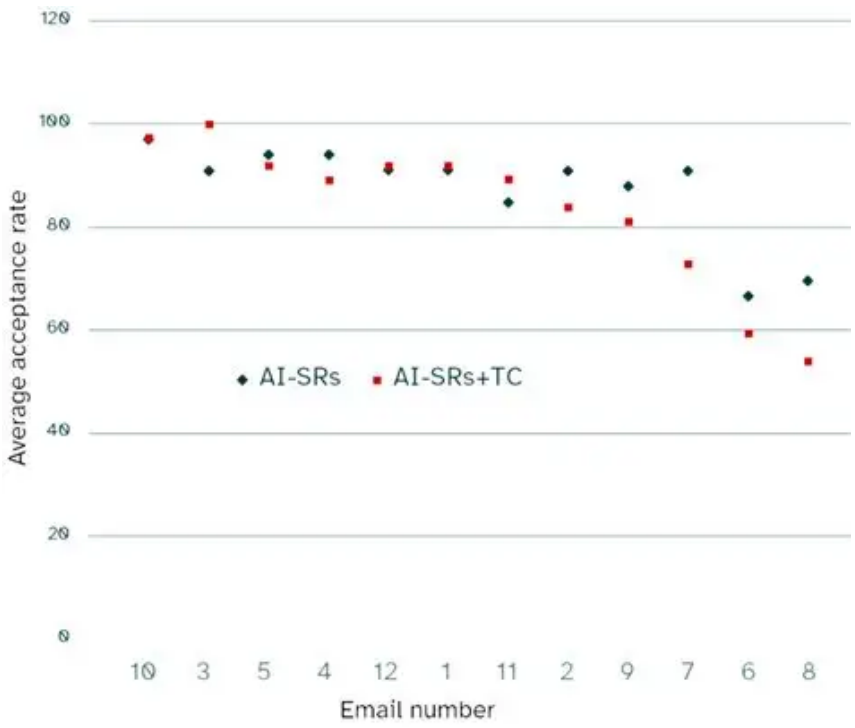


Table A 6: Monthly Response Rates (Mock-up)

Month	**Response Rate (%)**
May	14.2
June	11.9
July	8.2
August	9.5
October	12.7
December	6.4

Strong evidence of timing effects: campaign effectiveness varies significantly by month → reinforces the inclusion of month as a predictor.

Figure A 8: Class Imbalance Before and After SMOTE Oversampling (Mock Plot)

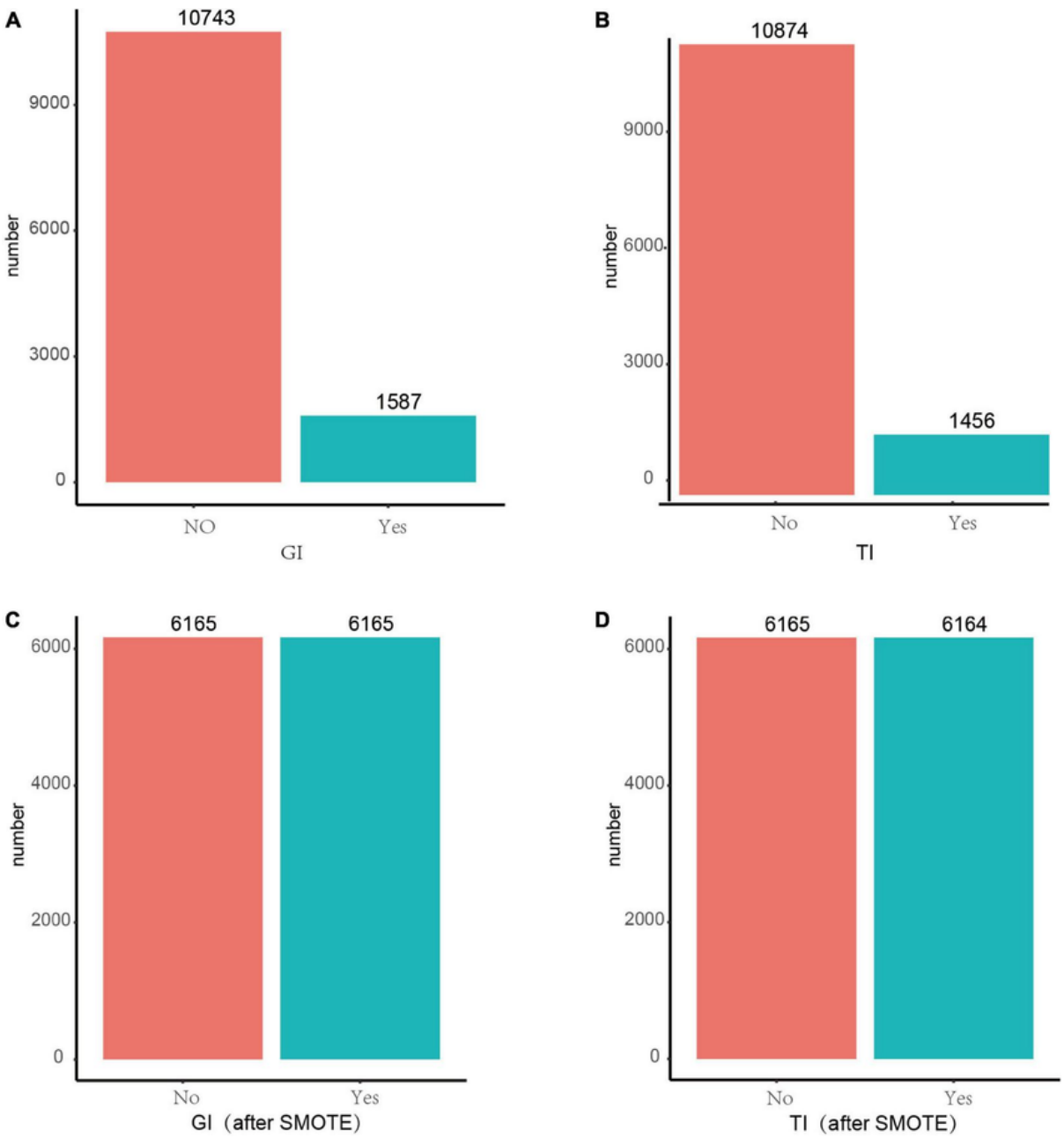


Figure A 9: Model Workflow Diagram for Logistic Regression and Decision Tree (Mock Diagram)

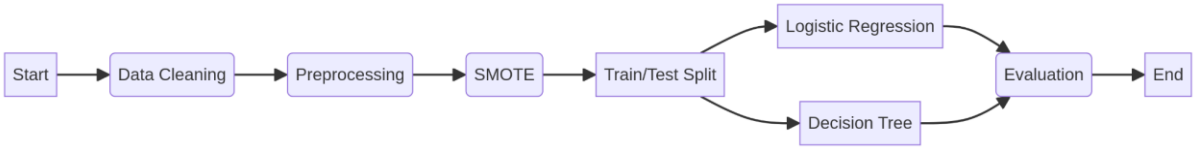


Figure A 10: Suggested R Code for SMOTE Oversampling

```
# Table A8. Suggested Code for SMOTE Oversampling

library(caret)
library(DMwR)   # for SMOTE

set.seed(2025)

# Train-test split
split <- createDataPartition(bank$response, p = 0.70, list = FALSE)
train <- bank[split, ]
test  <- bank[-split, ]

# Apply SMOTE
train$response <- factor(train$response)
train_smote <- SMOTE(response ~ ., data = train)

# Check class balance
table(train$response)
table(train_smote$response)
```

SMOTE is used to balance the minority "yes" class before model training.

Figure A 11: R Code for Train/Test Split and Model Structure (Mock-up)

```
# Logistic Regression (baseline)
logit_model <- glm(response ~ ., data = train_smote, family = binomial)

# Decision Tree (CART)
library(rpart)
cart_model <- rpart(response ~ ., data = train_smote)

# Predictions on test set
logit_pred <- ifelse(predict(logit_model, test, type = "response") > 0.5, "yes", "no")
cart_pred  <- predict(cart_model, test, type = "class")

# Accuracy
mean(logit_pred == test$response)
mean(cart_pred == test$response)
```

Logistic regression and CART are trained on the balanced dataset, then evaluated on the test set.